Muhammed Yousuf S

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Environmental Engineer | Air Quality | GIS | Environmental Data Analytics

Summary

Environmental Engineer with expertise in air quality monitoring, aerosol characterization, environmental data analytics, GIS, and source apportionment. Experienced in operating advanced environmental monitoring systems, developing analytical methods, processing complex environmental datasets, and delivering data-driven solutions for air quality assessment and environmental management. Proficient in Python, GIS, QA/QC, statistical analysis, hydraulic network modelling, and technical reporting.

Selected Technical Expertise

- Air quality monitoring and aerosol characterization
- Environmental analytical instrumentation (online & offline)
- Source apportionment (PMF) and trajectory modelling (HYSPLIT)
- Environmental data analytics, QA/QC, and statistical analysis
- GIS, remote sensing, and spatial analysis
- Python programming and scientific computing
- Hydraulic network modelling (WaterGEMS)
- Technical reporting and environmental assessment

Education

2023 - Present	Ph.D. in Environmental Engineering Civil and Environmental Engineering Indian Institute of Technology Delhi, India	CGPA: 9.25/10
2020 - 22	Master of Technology Civil Engineering APJ Abdul Kalam Technological University, College of Engineering Trivandrum, India	CGPA: 9.49/10
2019 - 20	Diploma in Civil Quality Assurance & Quality Control (QA/QC) Engineering National Council for Technology and Training, India	A+
2019 - 20	Quantity Surveying National Council for Technology and Training, India	A+
2015 - 19	Bachelor of Technology Civil Engineering APJ Abdul Kalam Technological University, India	CGPA: 7.53/10

Professional Experience

Indian Institute of Technology Delhi, India **2023–Present**
Doctoral Researcher

- Conducted ambient air quality monitoring using online and offline analytical instruments for comprehensive aerosol characterisation.
- Developed and optimized TD–GC/MS analytical methods for identification and quantification of organic molecular markers in atmospheric particulate matter.

- Performed integrated chemical characterization of atmospheric aerosols through analysis of organic, ionic, elemental, and carbonaceous (OC/EC) components.
- Applied source apportionment (PMF), atmospheric trajectory analysis (HYSPLIT), and statistical analysis to evaluate emission sources and atmospheric transport processes.
- Processed and analysed large environmental datasets using QA/QC protocols to generate technical reports and support environmental assessment and decision-making.

University of Nova Gorica, Slovenia, European Union
Indian Institute of Technology Delhi, India **2025**
Research Collaboration

- Conducted ambient aerosol characterisation using online and offline measurements to evaluate the physical, optical, and carbonaceous properties of atmospheric aerosols.
- Investigated measurement artefacts in aerosol absorption photometers and evaluated correction methodologies to improve the accuracy of aerosol absorption measurements.
- Analysed multi-instrument datasets from the SP2-XR, PTAAM-2λ, SMPS, Aethalometer, and Nephelometer to characterise aerosol optical properties, black carbon mixing state, and particle size distributions.
- Investigated the absorption enhancement associated with black carbon coating by integrating optical, microphysical, and carbonaceous aerosol measurements.
  Performed environmental data processing, QA/QC, statistical analysis, and interpretation of complex aerosol datasets to support air quality assessment and environmental monitoring.

Abel Engineering Consultants **2022**
GIS Engineer

- Performed advanced spatial analysis, geoprocessing, and thematic mapping to support engineering and planning projects.
- Designed and modelled water distribution networks using WaterGEMS, including hydraulic analysis and network optimisation.
- Integrated GIS datasets with engineering and field survey data to improve asset management and decision-making.
  Conducted geospatial data processing, quality assurance, and map production for client reports and technical documentation.

APJ Abdul Kalam Technological University **2020-2022**
College of Engineering Trivandrum
M.Tech Researcher

- Assessed spatial and temporal variations in water quality of Sasthamkotta Lake using geospatial techniques.
- Applied GIS and remote sensing tools for water quality modelling and mapping.
- Integrated satellite data and field-based observations for environmental analysis.
- Developed thematic maps to identify pollution patterns and critical zones.
  Performed data processing, analysis, and interpretation for environmental monitoring

Technical Skills

Instrumentation

- Thermal Desorption Gas Chromatography-Mass Spectrometry (TD-GC/MS)
- Organic Carbon/Elemental Carbon Analyzer (OC/EC Analyzer)
- Ion Chromatography (IC)
- Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES)

- Aethalometer
- Nephelometer
- Photothermal Aerosol Absorption Monitor (PTAAM-2λ)
- Single Particle Soot Photometer (SP2-XR)
- Scanning Mobility Particle Sizer (SMPS)
- Monitor for AeRosols and Gases in Ambient Air (MARGA)
- Ambient Metals Monitor (Xact)
- Micro-Orifice Uniform Deposit Impactor (MOUDI)

Software, Modelling & Data Analysis

- PMF
- HYSPLIT
- GIS
- WaterGEMS
- Python
- Origin
- Statistical Analysis
- QA/QC
- Environmental Data Processing

Journal Publications

- **Yousuf, S. M.**, Karigowda, Habib, G., Warsi, A. B., Imran, M., & Nabi, M. un. (2026). Chemical investigation of polycyclic aromatic hydrocarbon sources and associated health risks in PM2.5 from Eastern India. *Scientific Reports*. https://doi.org/10.1038/s41598-026-41899-0
- Shibu, K., Drisiya, J., & **Yousuf, S. M.** (2025). Spatiotemporal Modelling Approach for Nutrient Export in Sasthamkotta Freshwater Wetland Watershed. Journal of the Indian Society of Remote Sensing, 53(1), 225–239. https://doi.org/10.1007/s12524-024-01978-z
- Shibu, K., Ayoob, S., & **Yousuf, S. M.** (2023). Determination of land use–land cover changes in the adjoining panchayats of Sasthamkotta freshwater wetland, a Ramsar site in South India. International Journal of Lakes and Rivers, 16(1), 17–33.
- Deepak, K. R., Sabu, P., **Yousuf, S. M.,** & Thottumkal, C. G. (2022). Urban flood susceptibility map of Chennai using GIS and random forest method. International Research Journal of Engineering and Technology, 9(10).

Research Presentations

Yousuf, S. M., Habib, G. Seasonal Variation, Ring Distribution, and Source Characteristics of $PM_{2.5}$-bound PAHs in Eastern India. Platform Presentation at Asian Aerosol Conference 1-4 December 2025

Workshops and Trainings

- **Advanced Aerosol Science Intensive Course** (View Certificate)
  Centre for Atmospheric Sciences (CAS), IIT Delhi and Institute for Atmospheric & Earth System Research (INAR), University of Helsinki, 19th – 29th November 2025
- **Minds-on Air quality, humaN healTH & climAte Nexus** (View Certificate)
  Central University of Jammu & Sindhu Central University, 26th February – 1st March 2024
- **Aerosol Science Intensive Course** (View Certificate)

Centre for Atmospheric Sciences (CAS), IIT Delhi and Institute for Atmospheric & Earth System Research (INAR), University of Helsinki, 5th - 15th February 2024

- **Workshop on source apportionment** (View Certificate)
  Paul Scherrer Institute (PSI) and the partners of the Clean Air Project India – Indian Institute of Technology Bombay, 15th & 16th December 2023.
- **Workshop on WebGIS Enabled Spatial Information System** (View Certificate)
  TPLC, Govt. Engineering College Barton Hill Trivandrum - Directorate of Technical Education, Government of Kerala, 8th – 18th February 2021.

Awards and Achievements

- **Graduate Aptitude Test in Engineering 2019**
  Civil Engineering (Score: 321)
- **Civil Prize 2019**
  National Council for Technology and Training
- **Graduate Aptitude Test in Engineering 2022**
  Geomatics (Score: 450)

Positions of Responsibility

Teaching Assistant, IIT Delhi (2024–2025)

Courses

- Advanced Air Pollution Laboratory
- Air Pollution and Control
- Environmental Science

Responsibilities

- Conducted laboratory sessions
- Assisted classroom teaching
- Prepared assignments and examinations
- Coordinated field visits
- Supported course delivery